4-8 OTHER INTERVAL ESTIMATES FOR A SINGLE SAMPLE

4-8 OTHER INTERVAL ESTIMATES FOR A SINGLE SAMPLE

4-8.1 Prediction Interval

- **Goal**: In specific scenarios, the focus is on **predicting** a single future observation of a random variable rather than estimating its population mean.
- **Likely Range**: We aim to find a range of values associated with this specific prediction, known as a **prediction interval** (PI).
- **CI vs. PI**: A Confidence Interval (CI) on the mean is not appropriate here because estimating where the average lies is a different statistical problem than predicting the value of an individual new unit.

- **Example Application**: Using the golf club data from Example 4-7, a reasonable point prediction for a newly purchased club is the sample average ($\bar{x}$) of the previously tested clubs.

- **Confidence Level**: We can establish a $100(1-\alpha)\%$ **prediction interval** to provide a range for this new, individual observation.

4-8.1 Prediction Interval

- **Point Prediction**: For a random sample from a normal population, the sample mean $\bar{x}$ serves as a reasonable point prediction for a future observation X_{n+1}.

- **Prediction Error**: The expected value of the prediction error ($X_{n+1} - \bar{X}$) is zero, ensuring an unbiased prediction.

- **Variance Calculation**: The variance of the prediction error is $\sigma^2(1 + 1/n)$. The "1" represents the variability of the new observation itself, while the "$1/n$" represents the uncertainty in estimating the population mean.

- **Distribution Derivation:**
  - Since the population is normal, the prediction error is also normally distributed.
  - Replacing the unknown population standard deviation σ with the sample standard deviation S transforms the standard normal Z-ratio into a T-ratio.
- **Result:** This T-ratio follows a t-distribution with $n - 1$ degrees of freedom, providing the mathematical basis for constructing a **prediction interval**.

Prediction Interval

A **Prediction Interval** (PI) is used to estimate the range within which a single future observation is expected to fall, based on previously collected data.

- **Definition: A 100(1 - α)% PI on a single future observation from a normal distribution** provides a range that captures the value of a new observation (X_{n+1}) with a specific confidence level.

$$\bar{x} - t_{\alpha/2,n-1}s\sqrt{1+\frac{1}{n}} \le X_{n+1} \le \bar{x} + t_{\alpha/2,n-1}s\sqrt{1+\frac{1}{n}} \qquad (4\text{-}77)$$

Comparison of Intervals

- **Width**: The PI for X_{n+1} will always be longer than the CI for μ because there is more variability associated with the prediction error for X_{n+1} than with the error of estimation for μ.

- **Source of Error**:

  - **PI**: Based on the difference between two random variables ($X_{n+1} - \bar{X}$).

  - **CI**: Based on the difference between one random variable and a constant ($\bar{X} - \mu$).

- **Large Sample Behavior ($n \to \infty$):**
  - The **CI** length reduces to zero as it converges on the true value of the mean.
  - The **PI** length approaches $2z_{\alpha/2}\sigma$, meaning uncertainty remains for future observations even when there is no need to estimate distribution parameters.

Sensitivity to Normality

- **Robustness of Mean Estimates**: Confidence Intervals (CIs) and hypothesis tests on the mean are relatively insensitive to the normality assumption.
- **Sensitivity of PIs**: In contrast, Prediction Intervals (PIs) are rather sensitive to the normality assumption because they focus on a single future value drawn at random rather than an average.

EXAMPLE 4-17 Golf Clubs

- **Objective:** Determine the likely range of values (Prediction Interval) for the coefficient of restitution (COR) of a single new golf club.
- **Data:** Sample size (n) = 15, sample mean ($\bar{x}$) = 0.83724, and standard deviation (s) = 0.0245571.

Solution

- **Assumption:** A normal probability plot confirms the data follows a normal distribution.
- **Prediction Interval (PI):** Used to predict a single future observation (X_{n+1}).

$$\bar{x} - t_{\alpha/2,n-1} \cdot s\sqrt{1+\frac{1}{n}} \leq X_{n+1} \leq \bar{x} + t_{\alpha/2,n-1} \cdot s\sqrt{1+\frac{1}{n}}$$

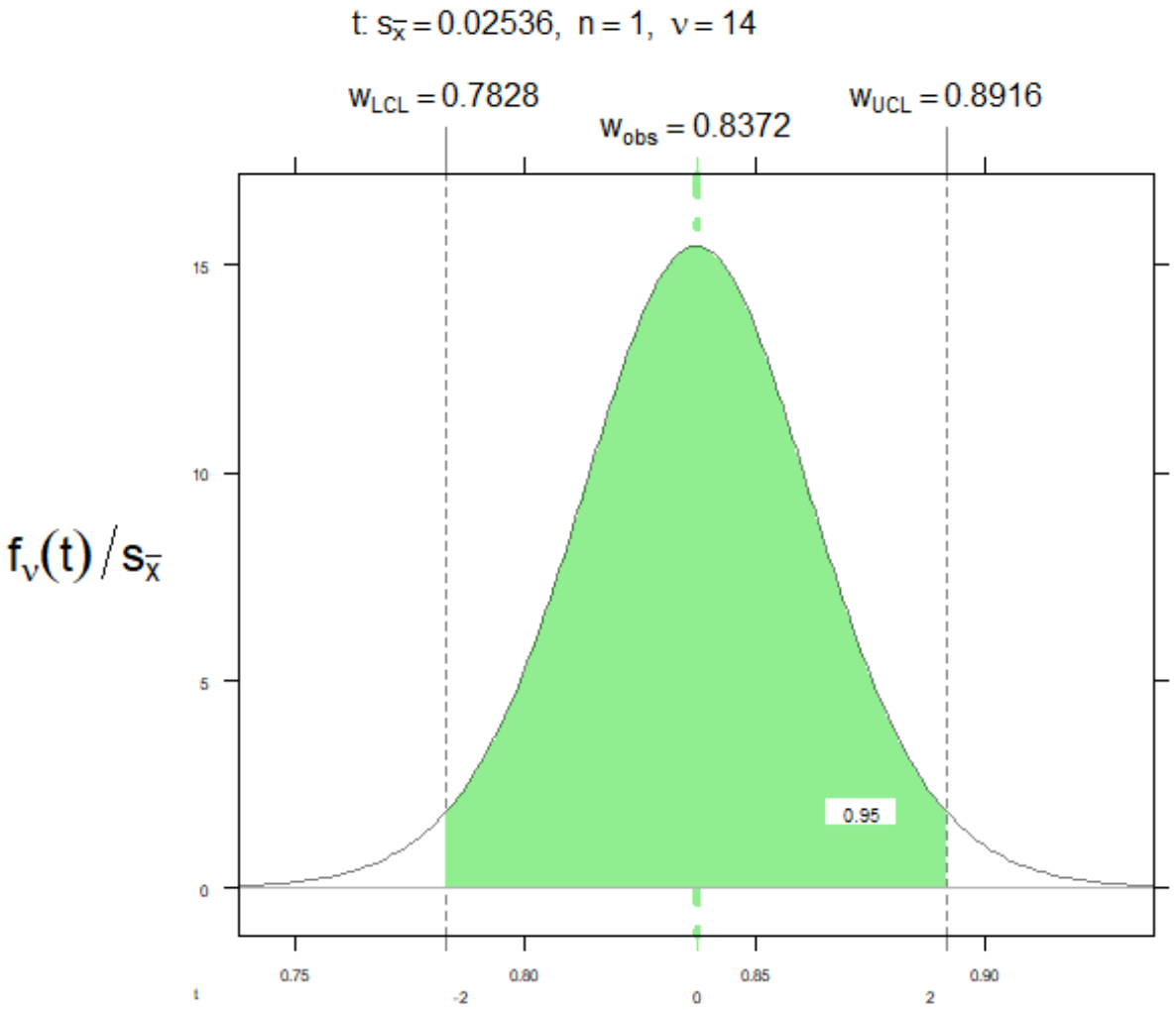
  - **Result:** $0.78284 \leq X_{16} \leq 0.89164$

- **Confidence Interval (CI):** Used to estimate the true population mean (μ).

$$\bar{x} - t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}$$

  - **Result:** $0.82364 \leq \mu \leq 0.85084$

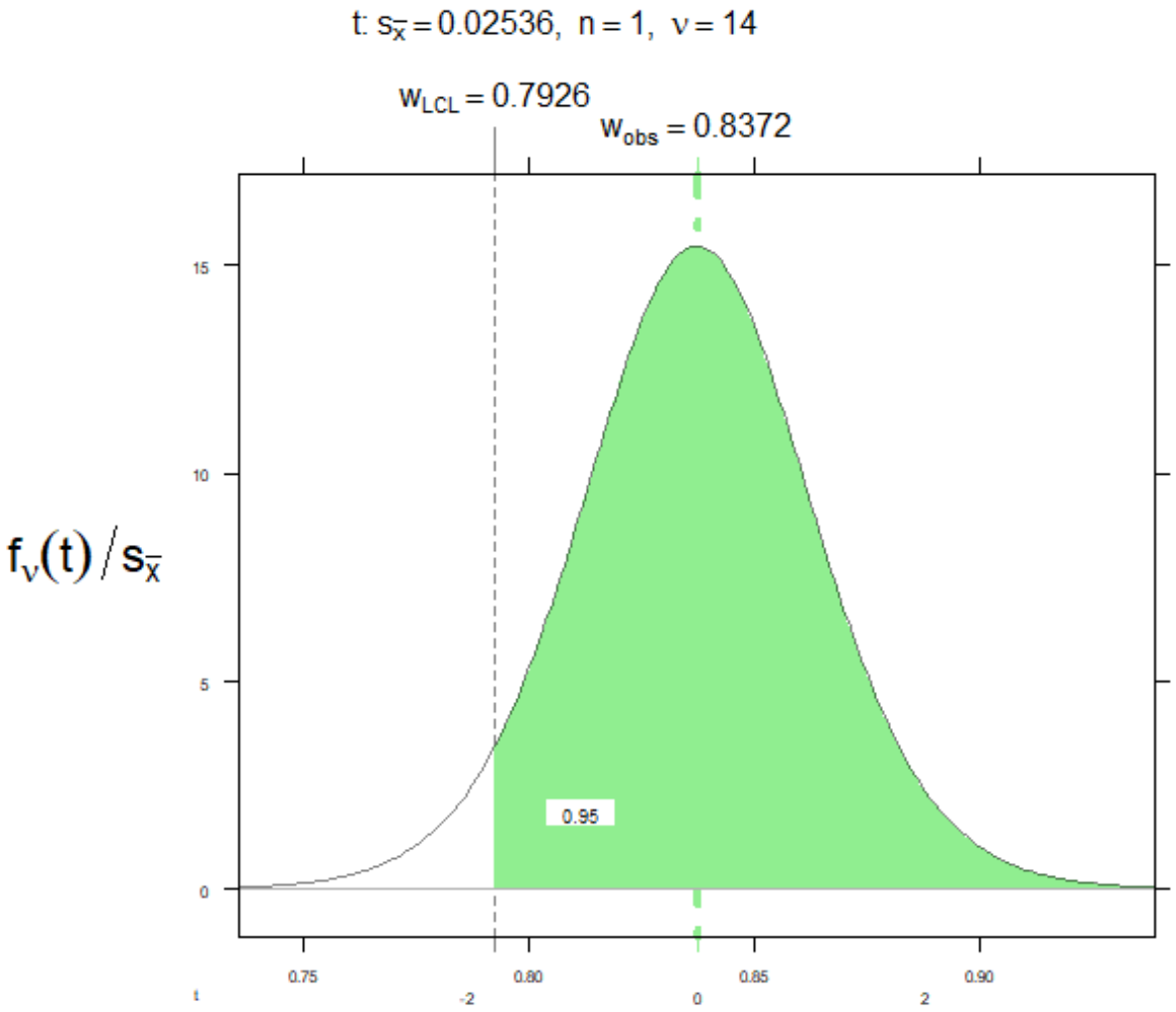
- **Key Takeaway:** The Prediction Interval is wider than the Confidence Interval because it must account for both the uncertainty of the mean and the variability of an individual data point.



One Sample t-test: COR.pred

	w_{obs}	w_{LCL}	w_{UCL}
$\bar{x}$	0.8372	0.7828	0.8916
t	0.0000	-2.1448	2.1448

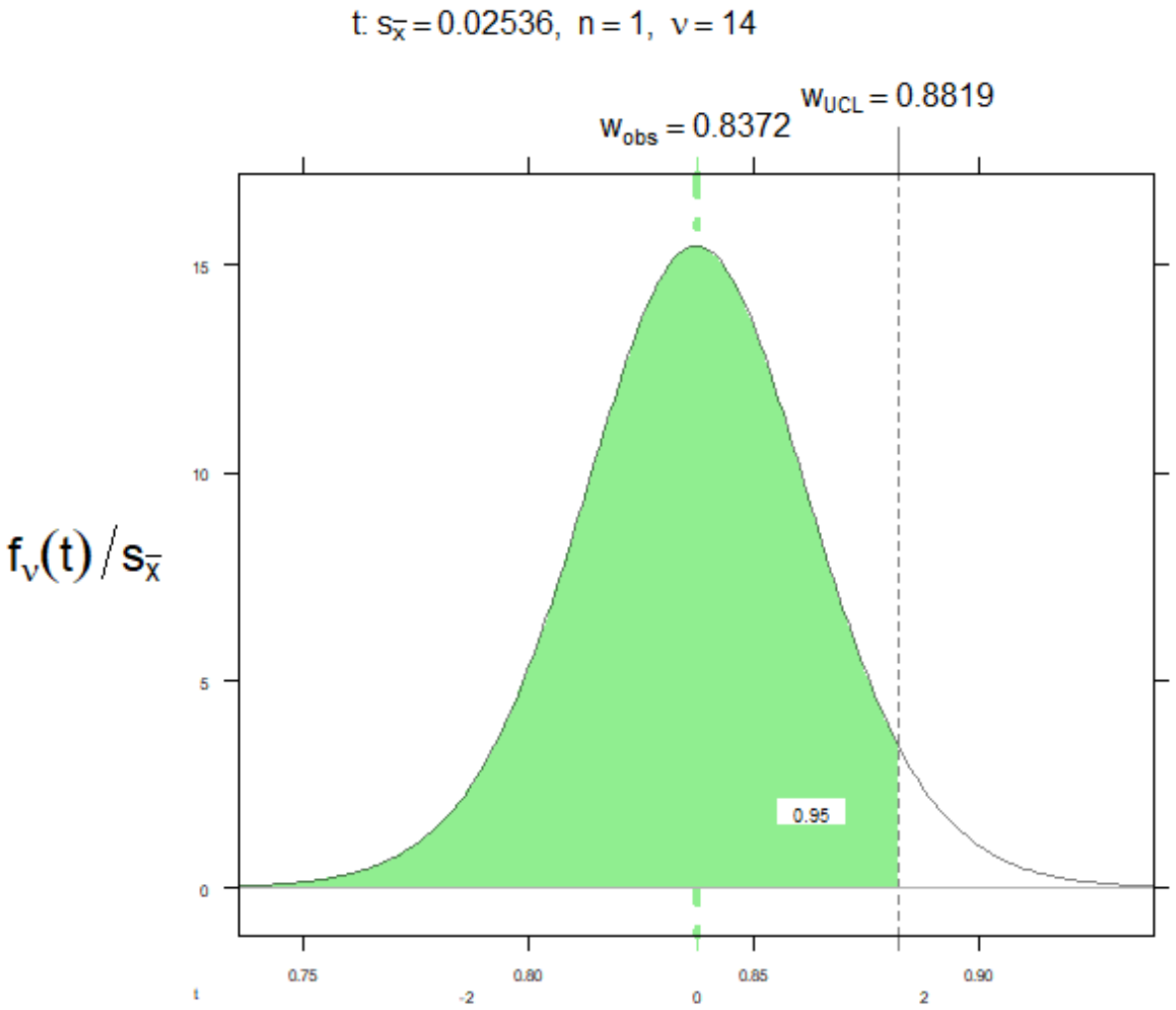
	Left	Confidence	Right
Probability	0.0250	0.9500	0.0250



One Sample t-test: COR.pred

	w_{obs}	w_{LCL}
$\bar{x}$	0.8372	0.7926
t	0.0000	-1.7613

	Probability
Confidence	0.9500



One Sample t-test: COR.pred

	w_{obs}	w_{UCL}
$\bar{x}$	0.8372	0.8819
t	0.0000	1.7613

	Probability
Confidence	0.9500

4-8.2 Tolerance Intervals for a Normal Distribution

- **Third Interval Type**: Beyond confidence and prediction intervals, a third type of interval is used to describe the distribution of the population itself.

- **Theoretical Basis**: If the population mean (μ) and standard deviation (σ) are known, an interval can be constructed to capture a specific percentage of the population (e.g., 95%) using the standard normal distribution.

- **Calculation Example**: With $\mu = 0.83$ and $\sigma = 0.025$, the range $[0.781, 0.879]$ captures 95% of the drivers, as ± 1.96 standard deviations covers 95% of the area under the normal curve.

4-8.2 Tolerance Intervals for a Normal Distribution

- **Definition**: Generally, an interval defined as $\mu \pm z_{\alpha/2}\sigma$ is called a **100(1 - α)% tolerance interval**.

- **Parameter Estimation**: When the population mean (μ) and standard deviation (σ) are unknown, they are estimated using the sample mean ($\bar{x}$) and sample standard deviation (s).

- **Sampling Variability**: Simply using a standard multiplier (like 1.96) with sample estimates $[\bar{x} \pm 1.96s]$ is insufficient, as the variability in the estimates often results in the interval capturing less than the target 95% of the population.

- **Correction Method**: To ensure the interval contains at least the desired proportion of the population, the standard z-multiplier is replaced with a larger calculated value.

4-8.2 Tolerance Intervals for a Normal Distribution

- **Confidence Requirement**: This replacement value allows us to state with a specific level of confidence that the interval covers at least the target percentage of the population.

4-8.2 Tolerance Intervals for a Normal Distribution

Tolerance Interval

A **tolerance interval** is used to capture a specific proportion of a population with a designated level of confidence.

- **Definition**: A **Tolerance Interval** designed to contain at least $\gamma\%$ of the values in a normal population with a confidence level of $100(1-\alpha)\%$ is given by:

$$[\bar{x} - ks, \bar{x} + ks]$$

- **Tolerance Factor (k)**: This factor is found in Appendix A Table VI and depends on the chosen confidence level ($1 - \alpha$) and the desired population proportion (γ).
- **Availability**: Values for k are commonly provided for:
  - Confidence levels ($1 - \alpha$) of 0.90, 0.95, and 0.99.
  - Population proportions (γ) of 90%, 95%, and 99%.
- **One-sided Bounds**: You can also compute one-sided **tolerance interval** factors using specific bounds also located in Appendix A Table VI.

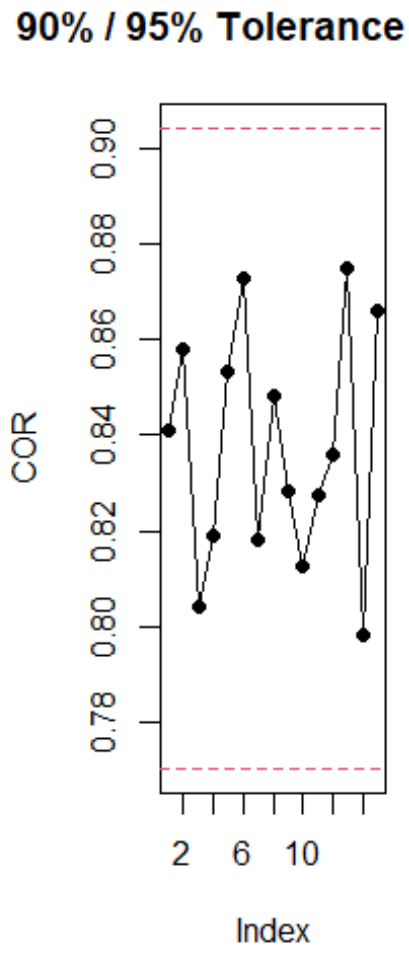
4-8.2 Tolerance Intervals for a Normal Distribution

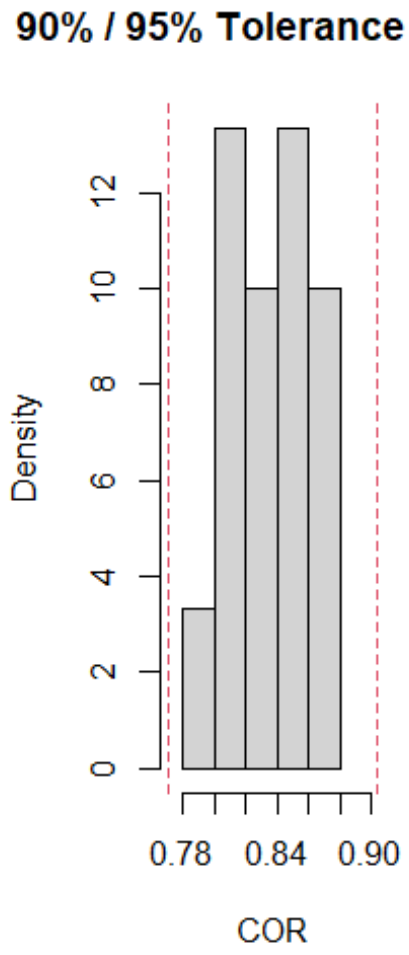
EXAMPLE 4-18 Golf Clubs

- **Objective:** Calculate a tolerance interval for the coefficient of restitution that covers at least 95% of the population with 90% confidence.
- **Data:** Sample size $n = 15$, mean $\bar{x} = 0.83724$, and standard deviation $s = 0.0245571$.
- **Factor:** The tolerance factor k is identified as 2.719553.

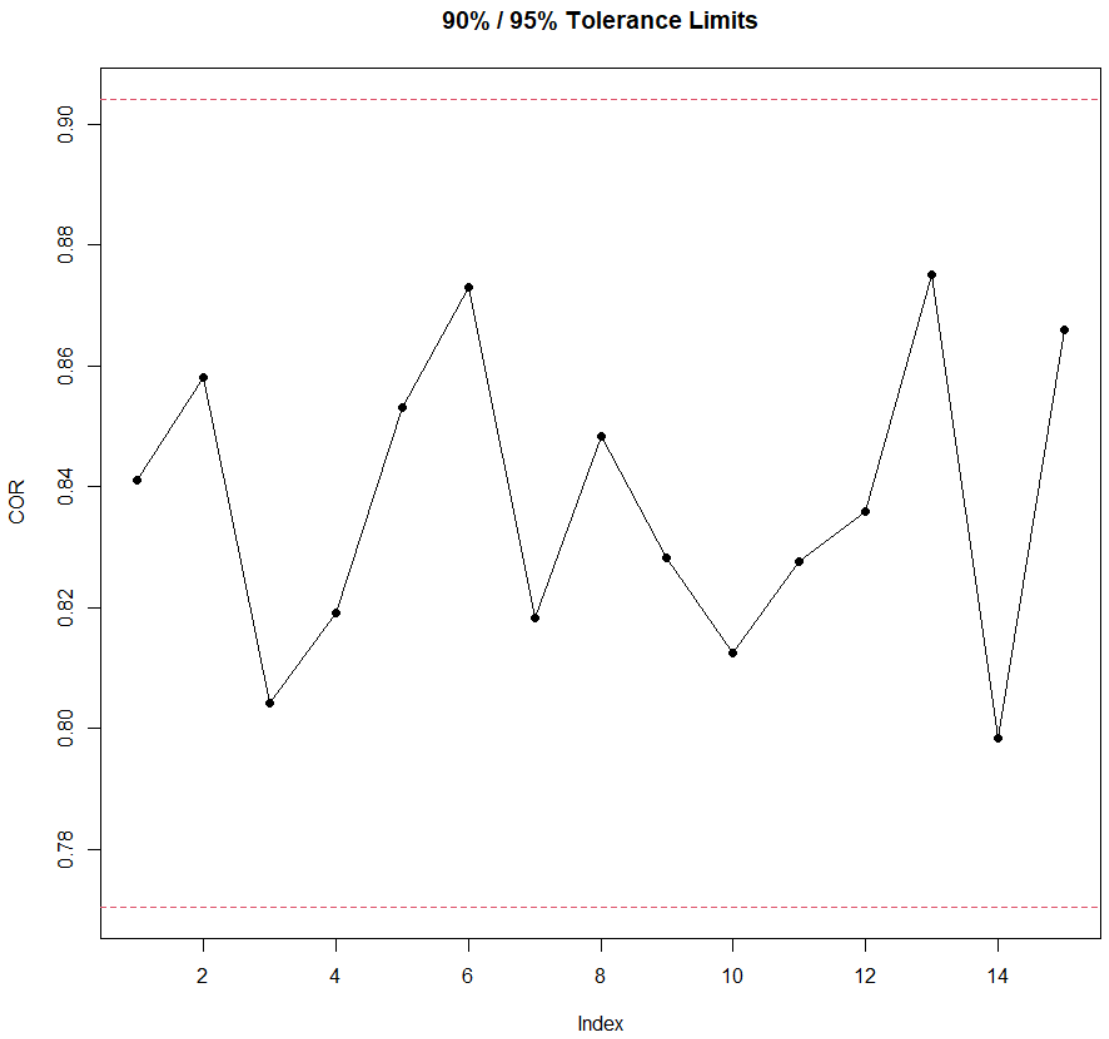
Solution:

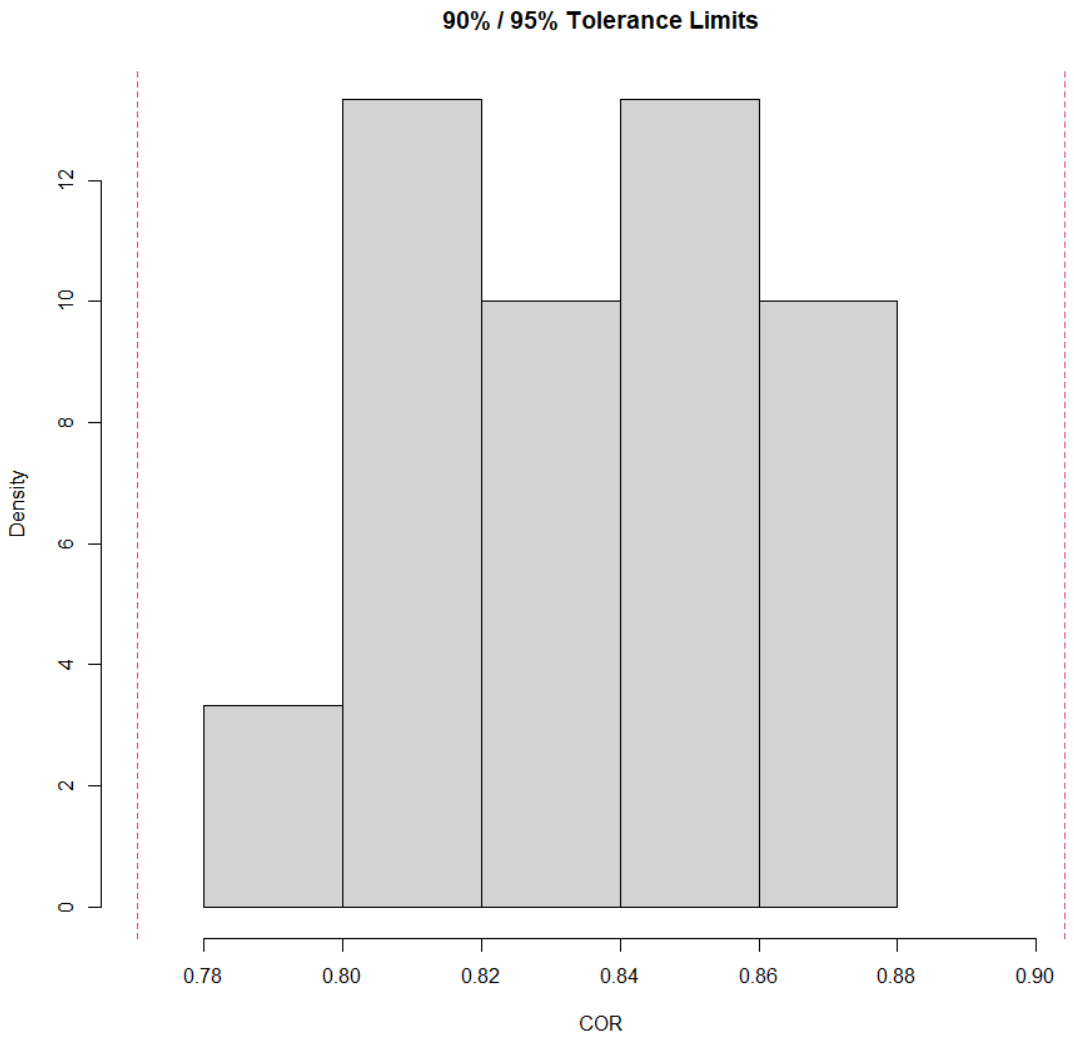
The interval is calculated using the formula $[\bar{x} - ks, \bar{x} + ks]$, resulting in a range of $[0.7704557, 0.9040243]$. This means there is 90% confidence that at least 95% of all clubs fall within these values.



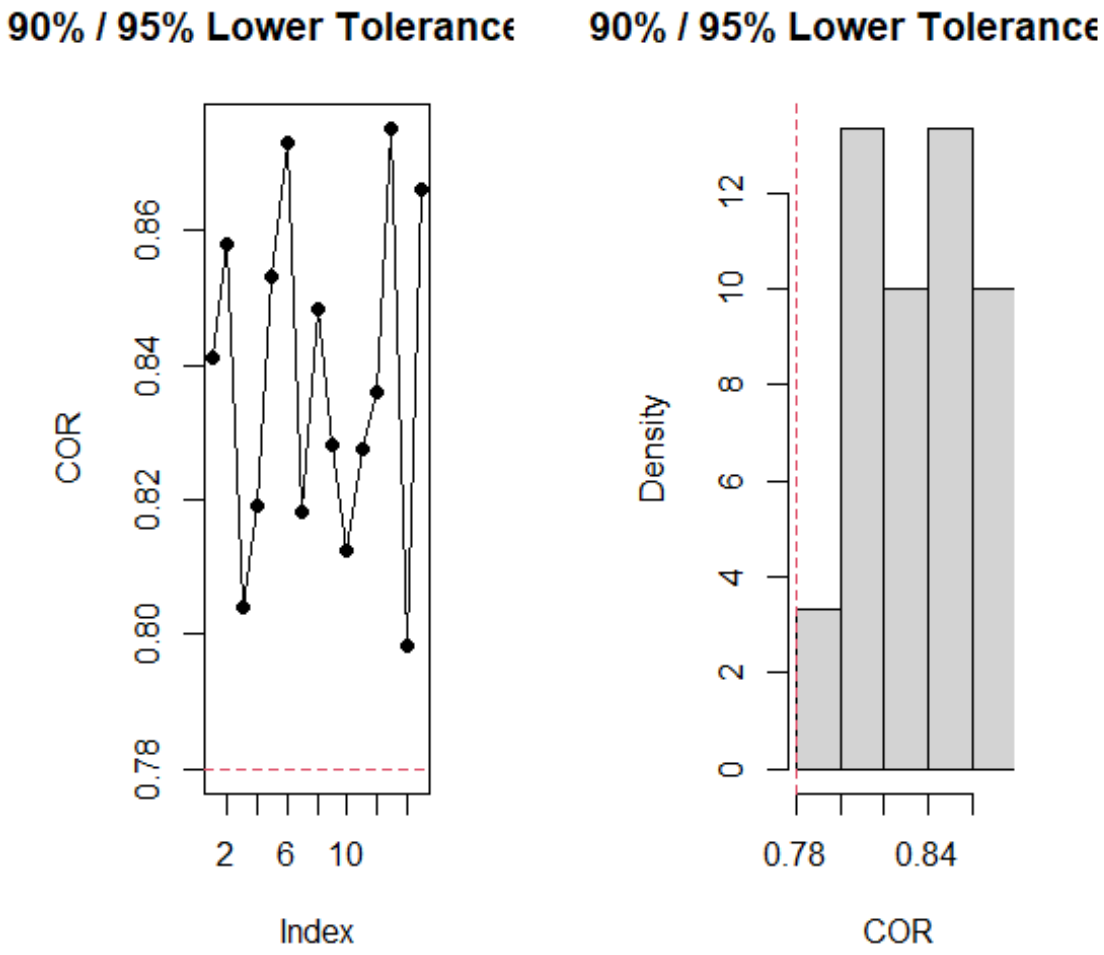


4-8.2 Tolerance Intervals for a Normal Distribution

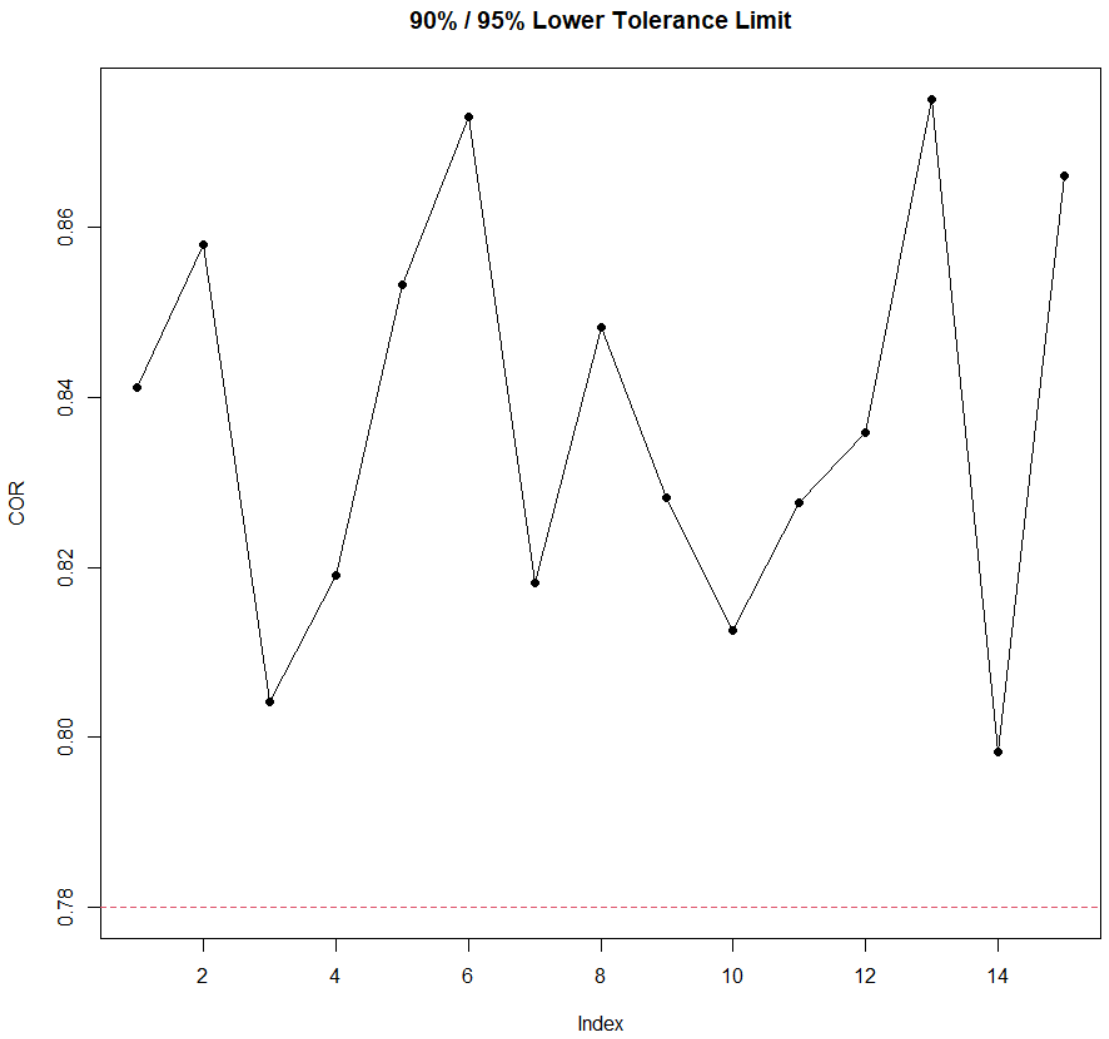


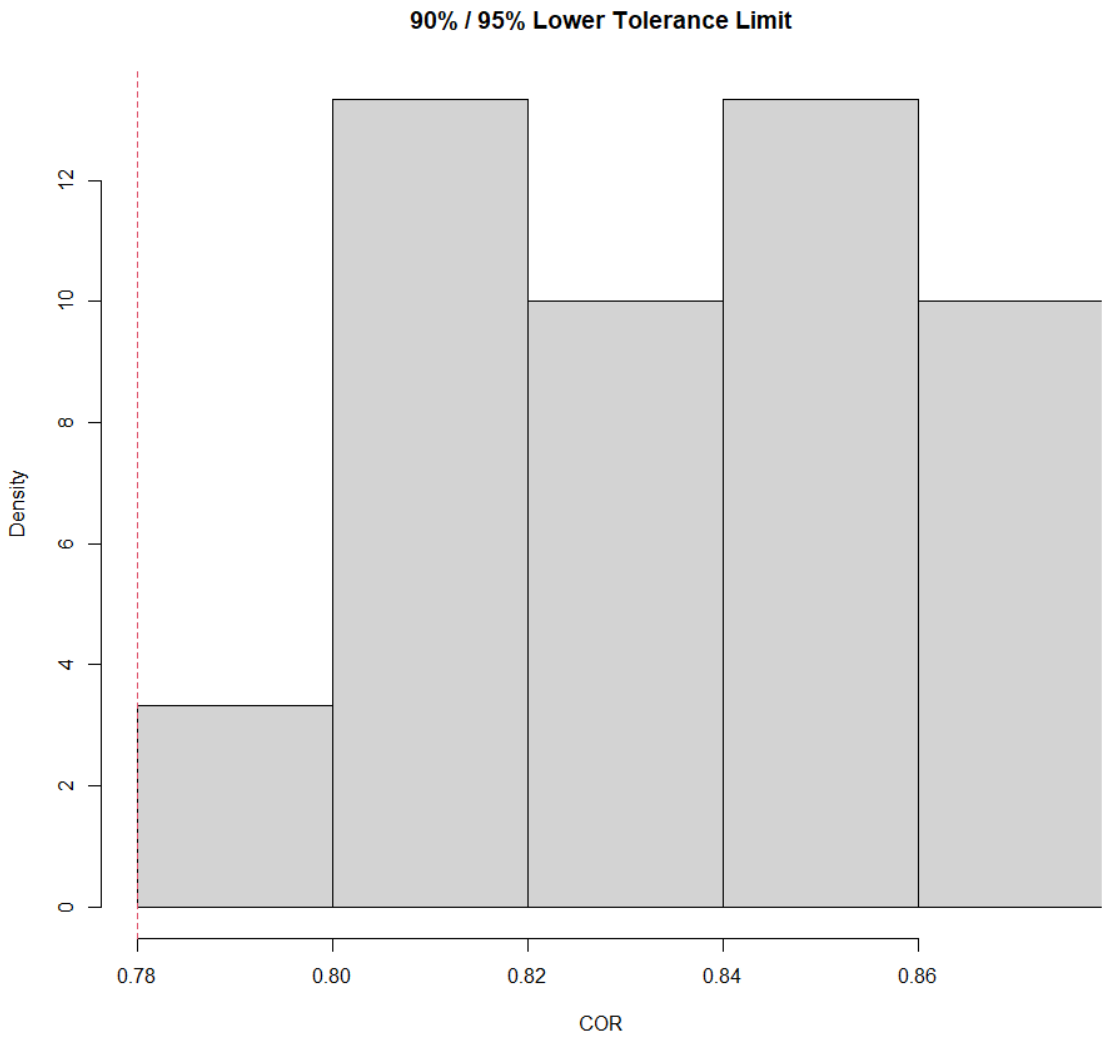


4-8.2 Tolerance Intervals for a Normal Distribution

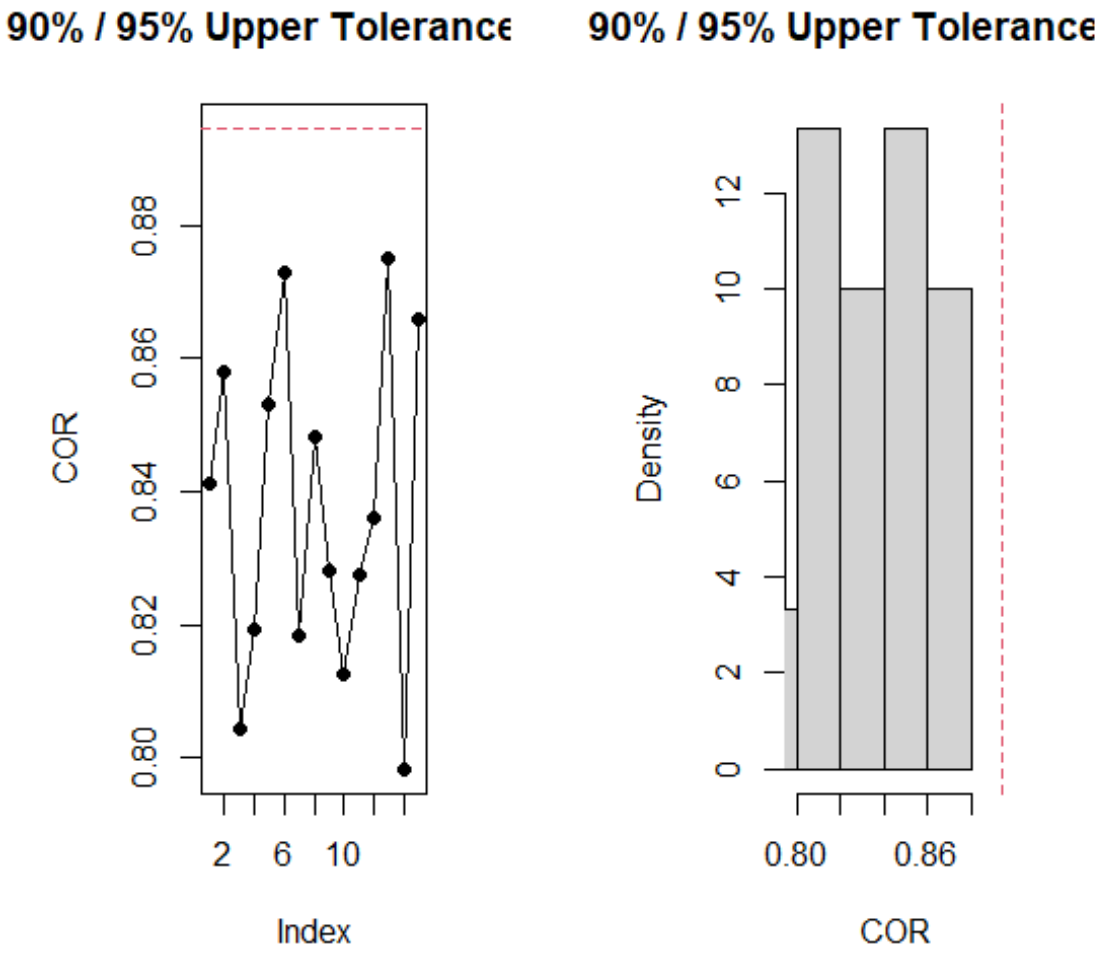


4-8.2 Tolerance Intervals for a Normal Distribution

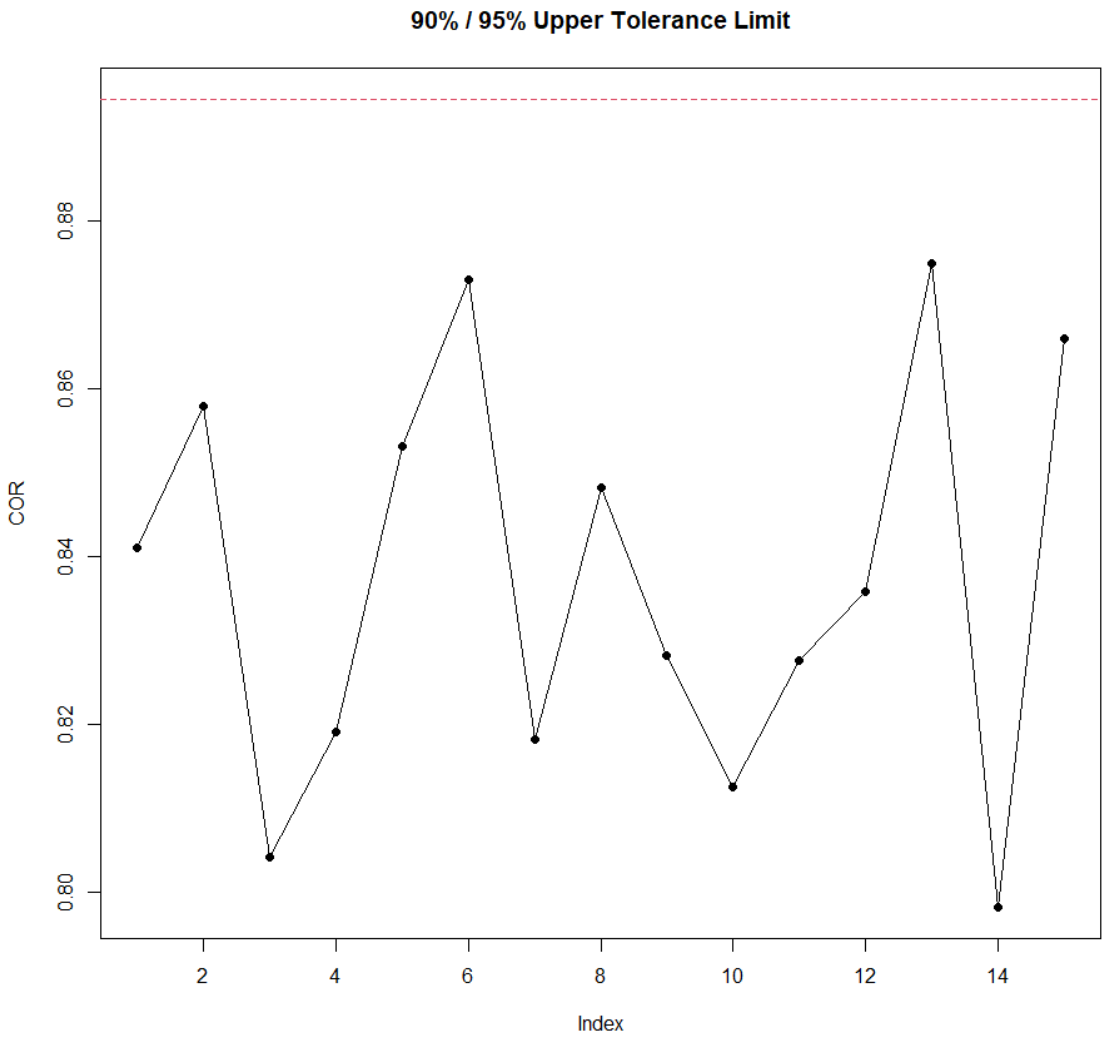


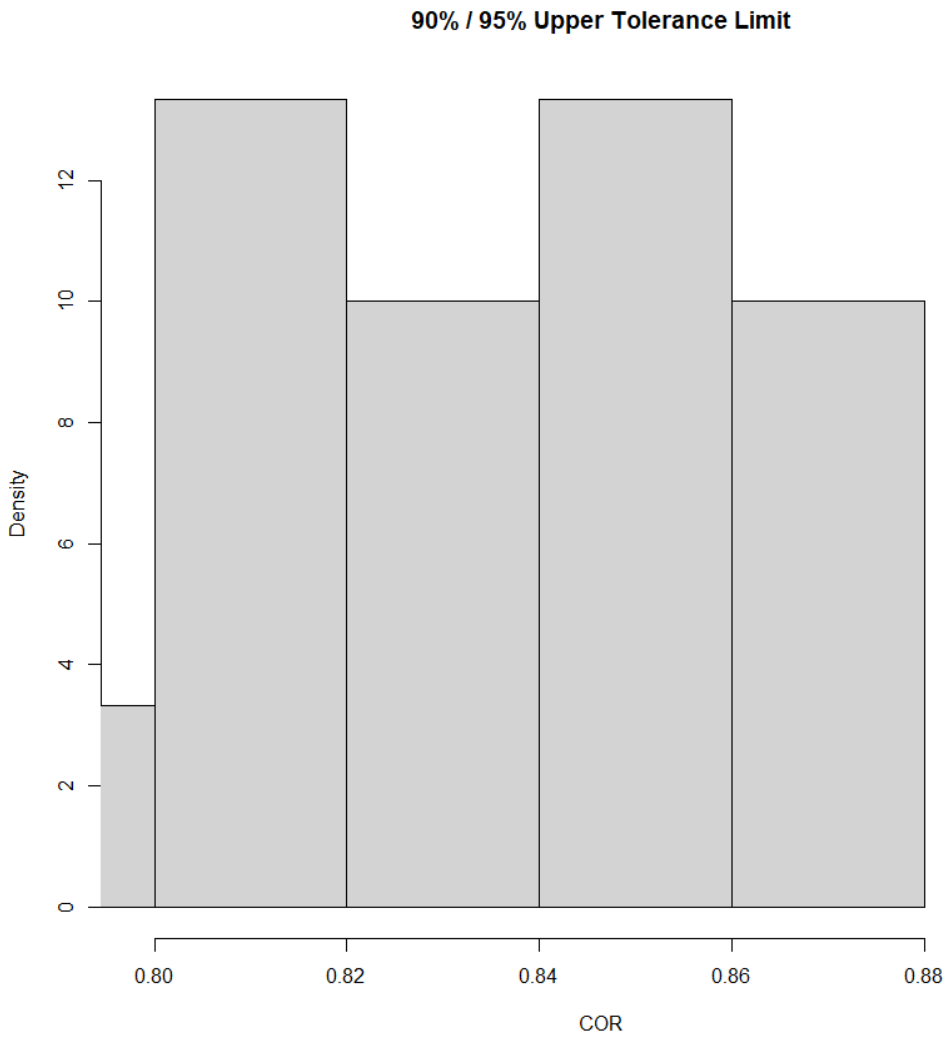


4-8.2 Tolerance Intervals for a Normal Distribution



4-8.2 Tolerance Intervals for a Normal Distribution





4-8.2 Tolerance Intervals for a Normal Distribution

Asymptotic Behavior of k

- **Convergence of k**: As sample size $n \to \infty$, the tolerance interval factor k approaches the z-value corresponding to the desired level of population containment.
  - **Example**: For 95% population containment, k approaches $z_{0.025} = 1.96$ (for two-sided intervals) as $n \to \infty$.
- **Relationship to PI**: As $n \to \infty$, a $100(1-\alpha)\%$ **prediction interval** on a single future observation converges to a **tolerance interval** that contains $100(1-\alpha)\%$ of the distribution.